



US012409355B2

(12) **United States Patent**
Thompson

(10) **Patent No.:** **US 12,409,355 B2**

(45) **Date of Patent:** **Sep. 9, 2025**

(54) **NON-FRANGIBLE THERMALLY
RESPONSIVE FLUID CONTROL
ASSEMBLIES FOR AUTOMATIC
CORROSION RESISTANT SPRINKLERS**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **17/974,020**

(22) Filed: **Oct. 26, 2022**

(65) **Prior Publication Data**

US 2023/0118207 A1 Apr. 20, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/636,889, filed as
application No. PCT/US2019/050751 on Sep. 12,
2019, now Pat. No. 11,484,739.
(Continued)

(51) **Int. Cl.**
A62C 37/12 (2006.01)
A62C 37/08 (2006.01)

(52) **U.S. Cl.**
CPC **A62C 37/12** (2013.01); **A62C 37/08**
(2013.01)

(58) **Field of Classification Search**
CPC .. B05B 1/02; B05B 12/08; B05B 3/04; B05B
1/262; B05B 1/20; B05B 1/265;
(Continued)

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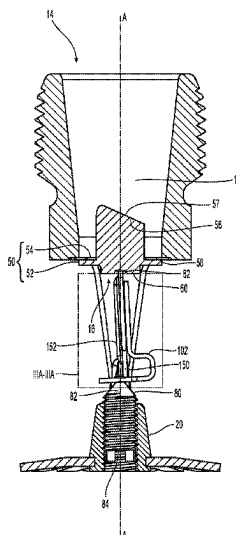
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(57) **ABSTRACT**

Corrosion resistant sprinklers and methods thereof include a
sprinkler frame body having an internal passageway with an
inlet and an outlet; a fluid deflecting member spaced from
the outlet and means for non-frangible thermal actuation
fluid control after exposure to an extreme salt environment.
Means include a seal assembly disposed in the outlet, a
screw member engaged with the sprinkler frame and a link
assembly in a supporting orientation with respect to the
sealing assembly to maintain and control transfer of a
sealing force of the screw member against the seal assembly
in a corrosive environment. Methods of obtaining or pro-
viding a corrosion resistant sprinkler include exposing sprin-
klers to a salt spray for an exposure period of over ten days
and maintaining seal integrity after the exposure period and
subsequently operating each sprinkler in a bath test within
3.5% of a nominal temperature rating of the sprinkler.

38 Claims, 9 Drawing Sheets



Related U.S. Application Data

- (60) Provisional application No. 62/800,020, filed on Feb. 1, 2019, provisional application No. 62/731,679, filed on Sep. 14, 2018.

(58) **Field of Classification Search**

CPC B05B 1/14; B05B 1/28; B05B 1/30; B05B 1/32; B05B 1/326; B05B 12/085; B05B 13/005; B05B 13/0278; B05B 15/65; B05B 3/06; B05B 3/14; B05B 7/1431; B05B 7/144; B05B 9/00; B05B 7/00; B05B 3/00; B05B 12/002; B05B 12/081; B05B 15/00; B05B 9/04; B05B 1/044; B05B 15/60; B05B 7/1481; A62C 37/12; A62C 35/68; A62C 37/11; A62C 31/02; A62C 37/08; A62C 35/58; A62C 37/50; A62C 35/10; A62C 35/13; A62C 37/14; A62C 37/04; A62C 2/242; A62C 99/009; A62C 3/08; A62C 35/023; A62C 5/033; A62C 99/0045

See application file for complete search history.

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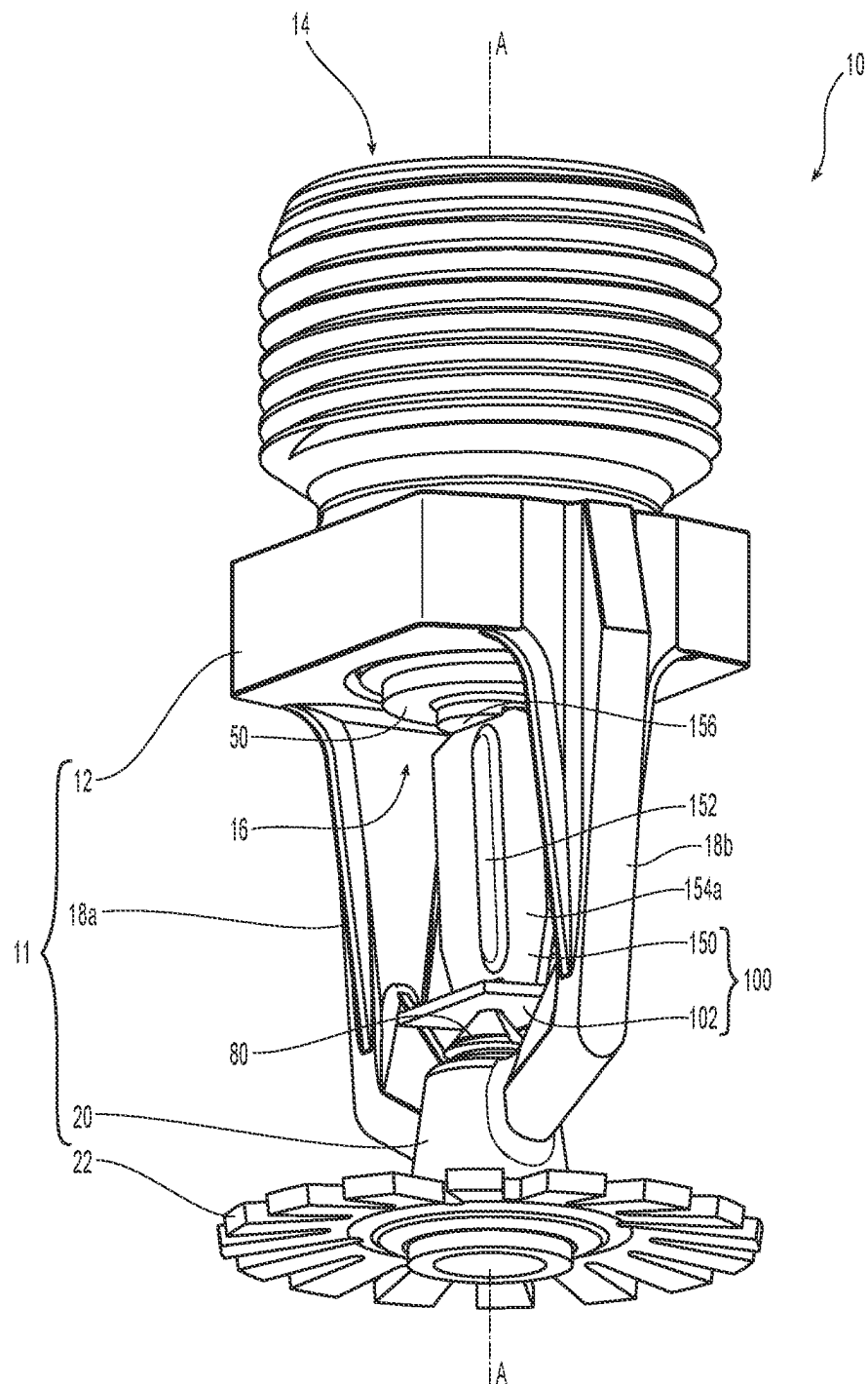


Fig. 1

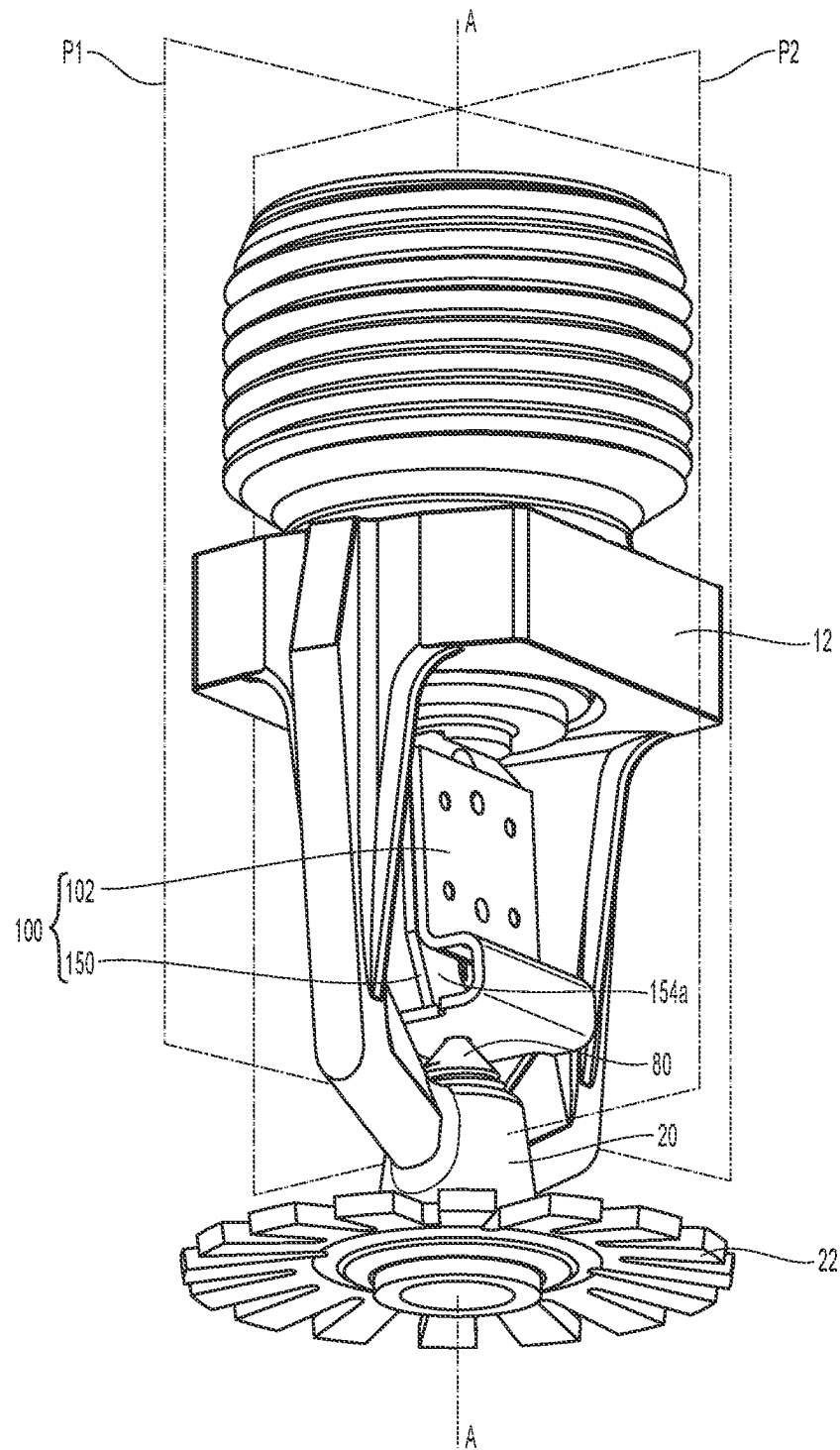


Fig. 2

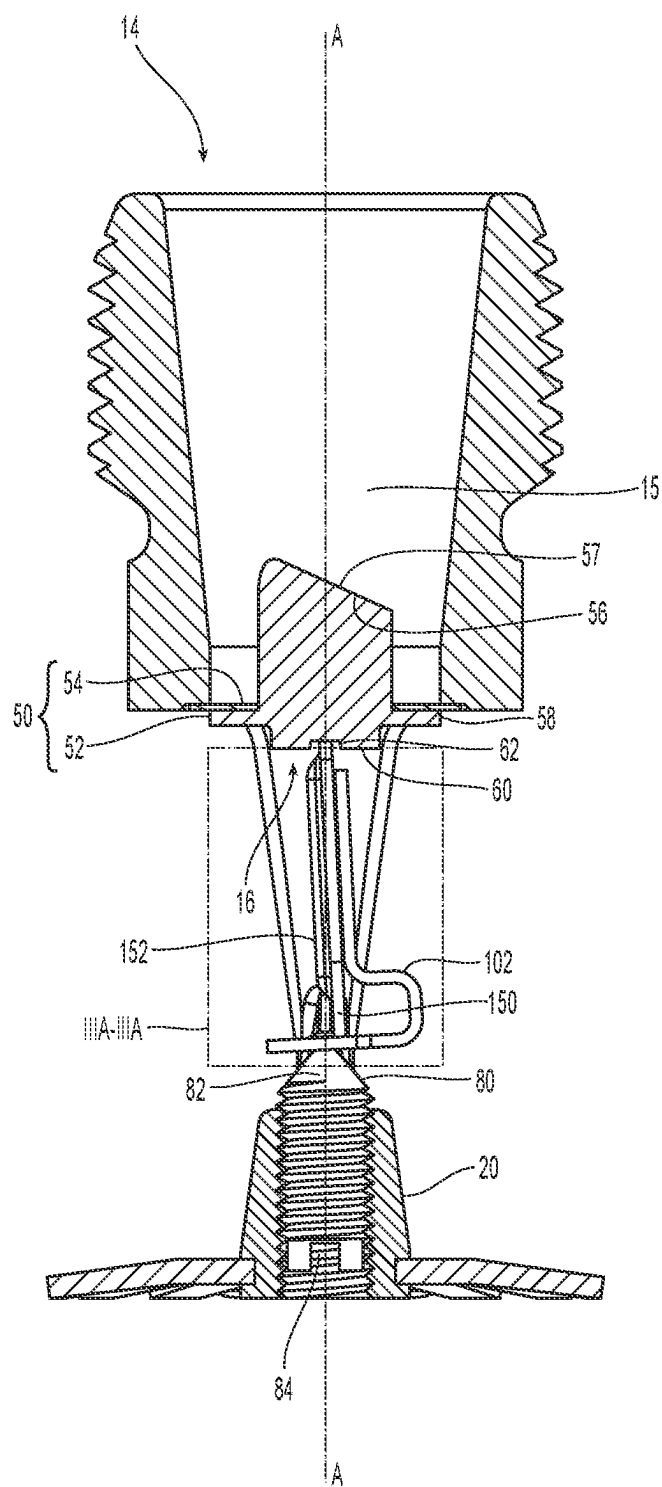


Fig. 3

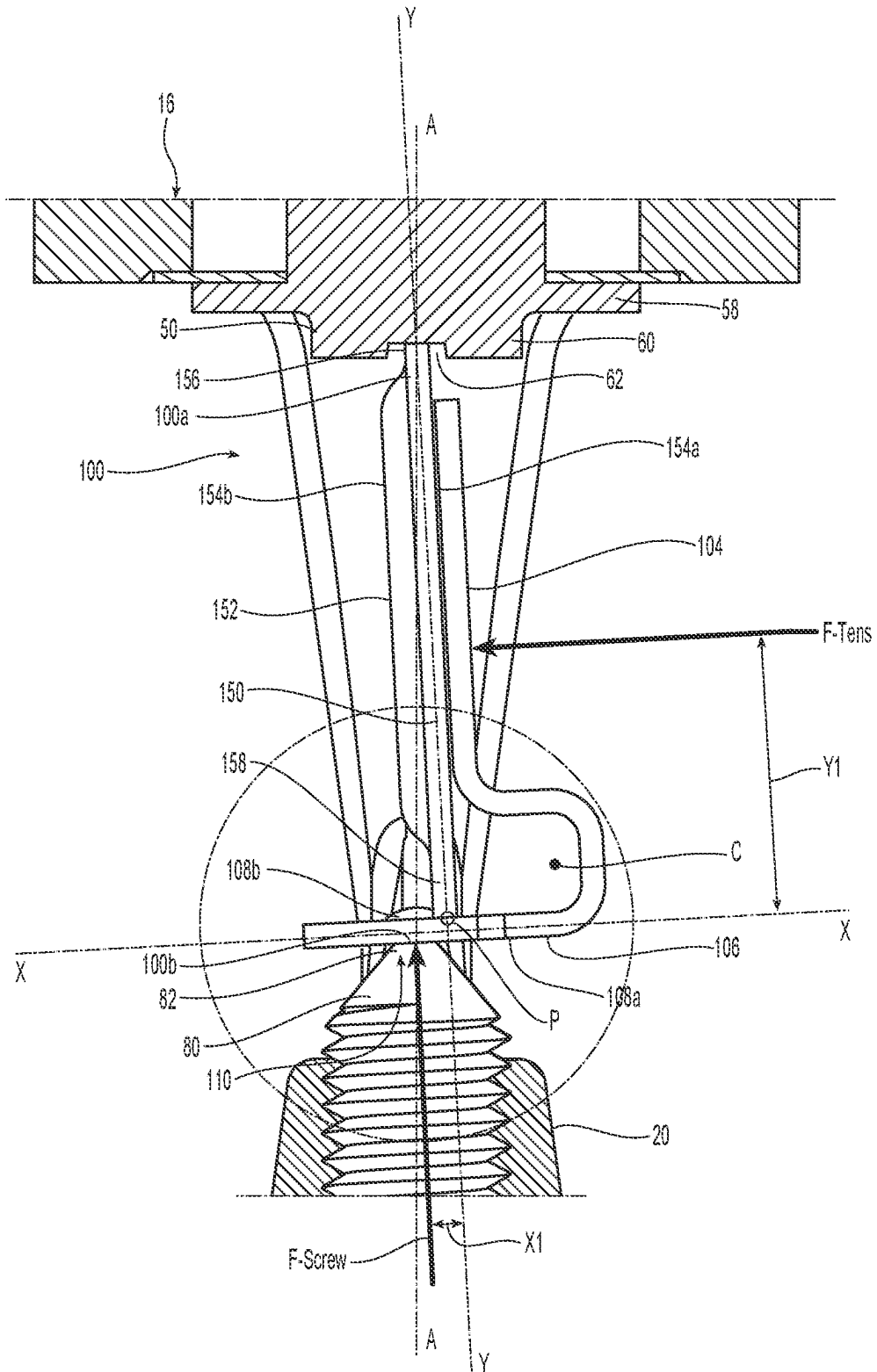


Fig. 3A

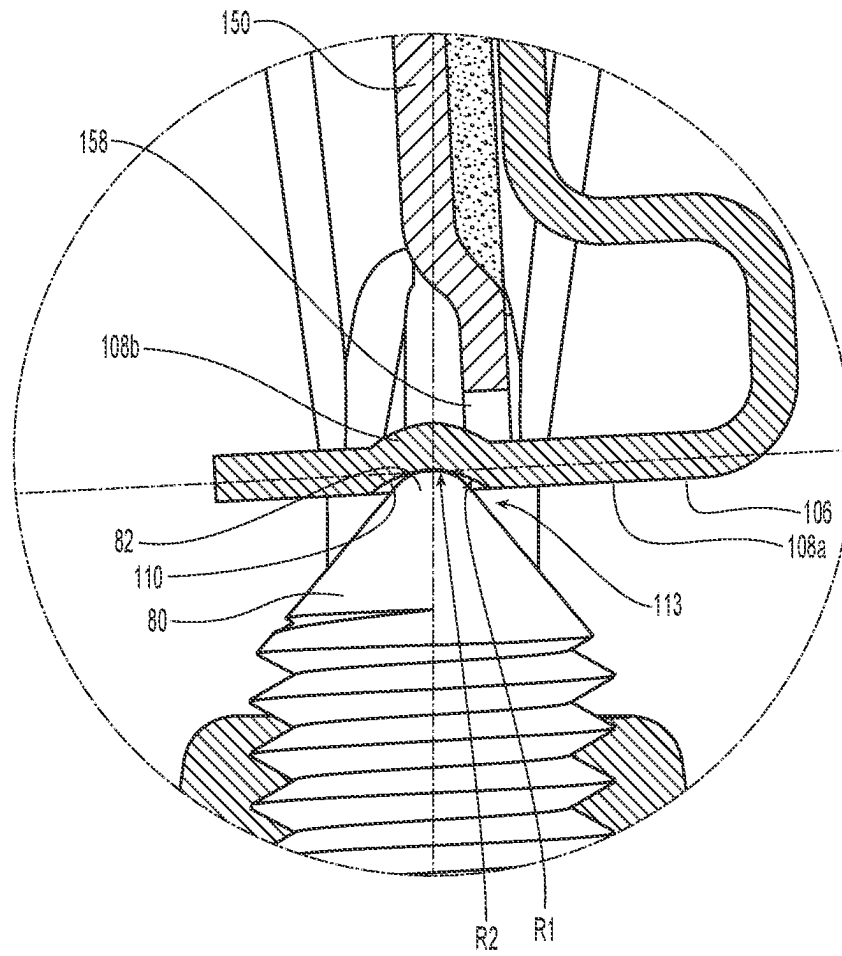


Fig. 3B

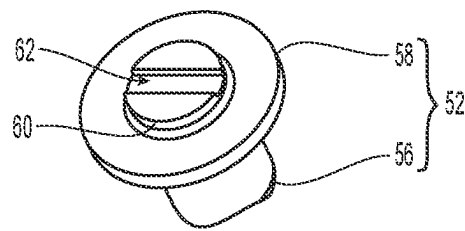


Fig. 4

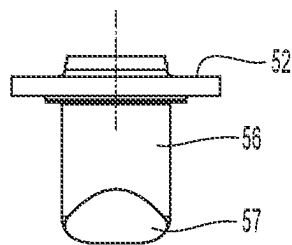


Fig. 4A

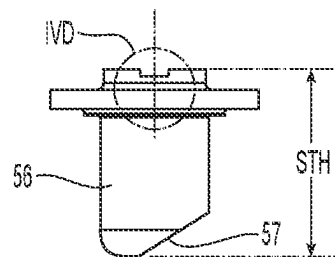


Fig. 4B

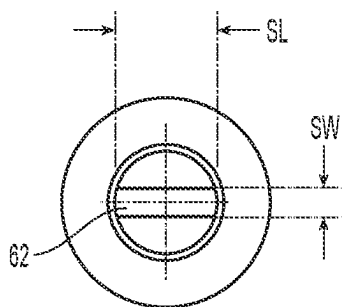


Fig. 4C

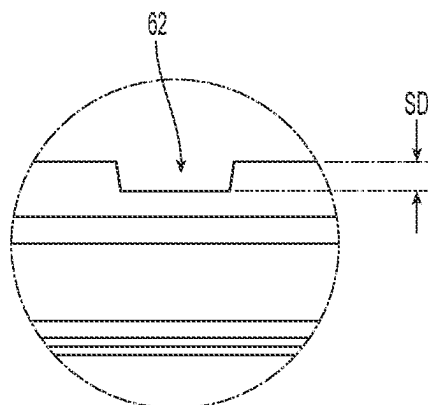


Fig. 4D

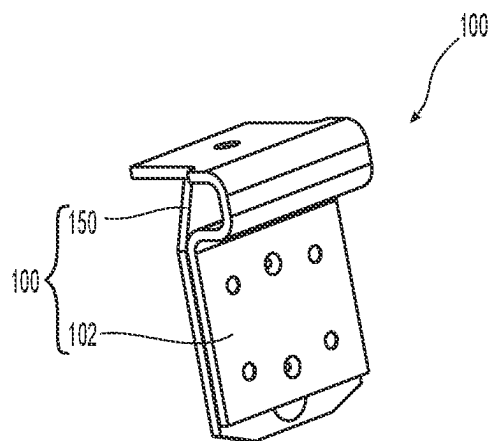


Fig. 5

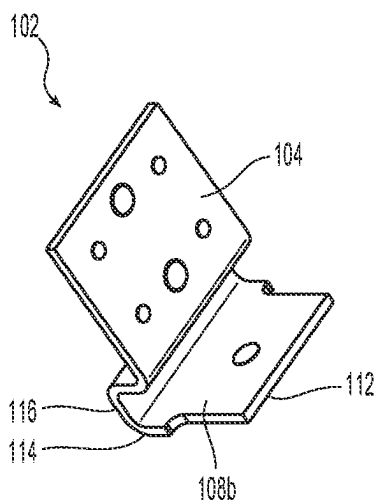


Fig. 5A

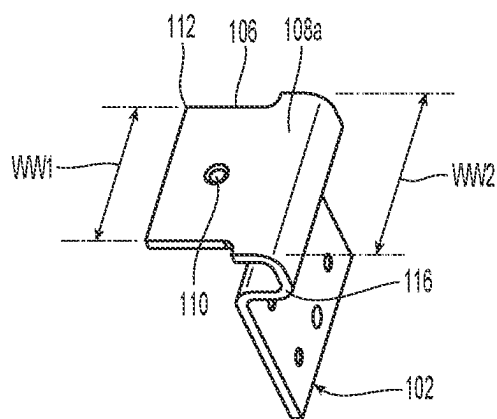


Fig. 5B

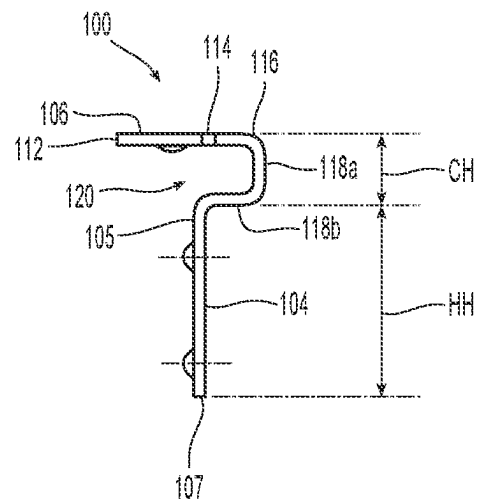


Fig. 5C

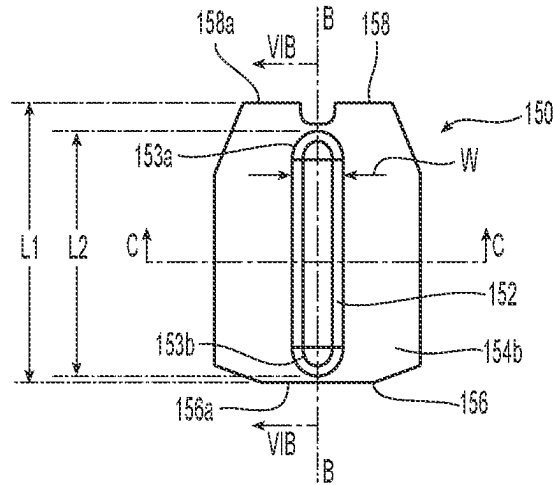


Fig. 6

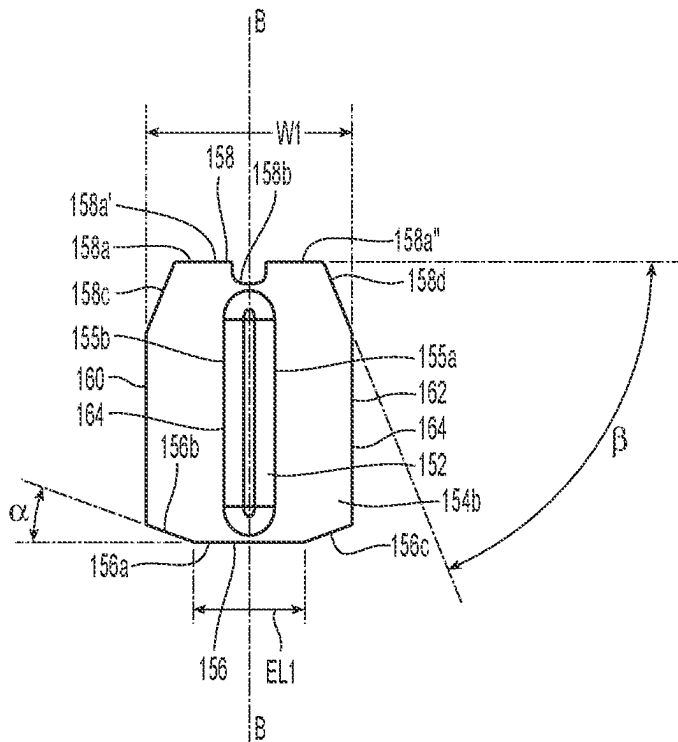


Fig. 6A

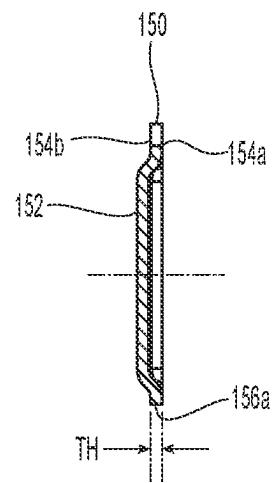


Fig. 6B

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**NON-FRANGIBLE THERMALLY
RESPONSIVE FLUID CONTROL
ASSEMBLIES FOR AUTOMATIC
CORROSION RESISTANT SPRINKLERS**

PRIORITY CLAIM & INCORPORATION BY
REFERENCE

This application is continuation application of U.S. patent application Ser. No. 16/636,889, filed Feb. 5, 2020, which is 35 U.S.C. § 371 application of International Application No. PCT/US2019/050751, filed Sep. 12, 2019, which claims the benefit of U.S. Provisional Application No. 62/731,679 filed Sep. 14, 2018 and U.S. Provisional Application No. 62/800,020 filed Feb. 1, 2019, each of which is incorporated by reference in its entirety.

TECHNICAL FIELD

The present invention relates generally to sealing and actuating assemblies for automatic sprinklers. More specifically, the present invention is directed to a non-frangible thermally responsive fluid control assembly for corrosion resistant sprinklers.

BACKGROUND ART

Generally, automatic fire protection sprinklers include a solid metal frame having a body and some type of deflector coupled to the frame to distribute fluid discharged from the body in a defined spray distribution pattern over an area to address a fire. Fluid discharge from an automatic fire protection sprinkler is automatically controlled by operation of a heat-responsive actuator that maintains a fluid tight seal at the discharge orifice by transferring a compressive sealing force on a cap (button or disc) or other sealing assembly from a load member such as, for example, a screw member engaged with the sprinkler frame that acts on the actuator. When the temperature surrounding the sprinkler is elevated to a pre-selected value indicative of a fire, the actuator operates and thereby ceases transfer of the sealing force, permitting ejection and release of the cap by the discharge of fluid through the unsealed sprinkler. There are generally two types of thermally responsive actuators: frangible and non-frangible. Frangible actuators generally include a liquid-filled frangible bulb that shatters upon reaching its rated temperature. Non-frangible actuators can include fusible links or soldered mechanical arrangements in which the components of the assembly separate upon fusion of the solder reaching its rated temperature. An exemplary fusible link is shown and described in U.S. Pat. No. 4,623,023.

Automatic sprinklers are used in a variety of environments including manufacturing facilities. In these facilities, the sprinklers may be installed in corrosive environments or exposed to corrosive conditions. Over time, corrosive effects can prevent or interfere with the proper sprinkler thermal response, ejection and/or fluid distribution. To combat the impact of corrosive effects on fire protection, there are corrosion resistant sprinklers. A “corrosion resistant sprinkler” is a sprinkler that is designed to resist exterior elements that attack a standard brass sprinkler. Corrosion resistant sprinklers are fabricated from corrosion resistant materials, such as for example, solid stainless steel. Fabricating sprinklers from such a material can provide for a sprinkler frame that is uncoated and thus free of a separate material for corrosion resistance. Such uncoated frames can be treated with a passivation process, such as for example, ASTM

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A967/A967-M-17: Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts.” Even without specific treatment, some alloys including stainless steel are self-passivating and thus provide a self-generating corrosion resistant coating. As used herein, self-passivating is a material property by which a base material becomes corrosion resistant by a microcoating that is created by oxygen in the atmosphere reacting with the base material. Corrosion resistant sprinklers can be made of a material that has a resistance to corrosion equal to or exceeding that of bronze alloy having a minimum copper content of 80 percent. A corrosion resistant sprinkler is distinguished from one that is “corrosion proof.” Corrosion proof is a sprinkler having a coating that can withstand corrosive effects. In order for a corrosion resistant sprinkler to be used in corrosive environments, such sprinklers are evaluated for their performance integrity under such conditions including their seal integrity and thermal responsiveness. Such integrity tests include evaluating the seal integrity of a corrosion resistant sprinkler after exposure to an extreme salt environment. As used herein an “extreme salt environment” is one at least equal to a misting environment having a sodium chloride concentration of 20% by weight.

Fire protection sprinklers sold and used as corrosion resistant sprinklers are generally subject to industry accepted fire code requirements and the approval of the “authority having jurisdiction” (AHJ) to ensure compliance with the applicable codes and requirements. One manner of satisfying the applicable requirements, is by identification of fire protection sprinklers capable of corrosion performance through appropriate industry accepted corrosion and operating testing. To facilitate the AHJ approval process, fire protection equipment can be “listed,” which as defined by NFPA 13, means that the equipment is included in a list by an organization that is acceptable to the AHJ and whose list states that the equipment “meets appropriate designated standards or has been tested and found suitable for a specified purpose.” One such listing organization includes, Underwriters Laboratories Inc. (“UL”). UL Standard for Safety for Automatic Sprinklers for Fire-Protection Service UL 199 (11th ed. 2005, rev. 2008) (“UL 199”) provides a corrosion and operating temperature test for corrosion resistant sprinklers. Similar testing standards for corrosion resistant sprinklers are provided by FM Approvals LLC (“FM”). Such FM Approvals include: (i) “Approval Standard for Automatic Sprinklers for Fire Protection-Class Number 2000” (February 2018) (“FM 2000”); (ii) “Approval Standard for Quick Response Storage Sprinklers for Fire Protection-Class Number 2008” (February 2018) (“FM 2008”); or (iii) “Approval Standard for Residential Sprinklers for Fire Protection-Class Number 2030” (August 2009) (“FM 2030”).

As part of the requisite performance testing for corrosion resistant sprinklers in an extreme salt environment, each of the UL and FM standards have a 30-Day corrosion test and an operating temperature bath test in addition to other corrosion resistant testing. Generally, the test sprinklers are exposed to a specific corrosive environment for 30 days to determine whether the sprinkler and its actuator can maintain its fluid tight seal. If the sprinkler satisfies the 30-day test, the sprinkler is subjected to a thermal sensitivity test and an operating temperature bath test. In the thermal sensitivity test, the tested sprinkler is exposed to an air stream at a controlled temperature and velocity and its actuation response is determined for evaluating sensitivity compliance. In the bath test, the tested sprinkler is exposed to a heated bath which is raised to the stated operating

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temperature to verify that the sprinkler and its actuator appropriately thermally responds and operates. The standards include other applicable corrosion resistance tests depending upon the material and/or configuration of the sprinkler. Additional corrosion resistance tests can include stress cracking testing, carbon dioxide-sulfur dioxide exposure testing, and/or hydrogen sulfide testing.

Although there are known corrosion resistant sprinklers, it is believed that there are no known corrosion resistant sprinklers with a thermally responsive non-frangible actuator that have passed both 30-day corrosion testing and operating temperature testing. Moreover, there are no known corrosion resistant sprinklers with a thermally responsive non-frangible actuator that have passed each of the 30-day corrosion testing, thermal sensitivity and operating temperature testing. For certain facilities, such as for example food processing plants, installing corrosion resistant sprinklers with frangible glass bulb-type actuators can create operational problems because of the difficulty in recovering and cleaning any glass remnants from bulbs that have been shattered either by thermal actuation or by inadvertent contact from equipment or personnel. Accordingly, there is a need for corrosion resistant sprinklers with a non-frangible thermally responsive actuator and seal assembly that can control fluid discharge and is capable of withstanding applicable corrosive testing. In particular, there is a need for corrosion resistant sprinklers with a non-frangible thermally responsive fluid control assembly that is capable of withstanding an extreme corrosive salt environment.

DISCLOSURE OF INVENTION

Preferred embodiments of an automatic corrosion resistant fire protection sprinkler are provided that preferably include a corrosion resistant frame and non-frangible thermally responsive fluid control assembly. Preferred embodiments of the non-frangible thermally responsive fluid control assembly include a seal assembly and a thermally rated link assembly disposed in a supporting orientation with respect to one another to maintain and control transfer of a sealing force against the seal assembly in a corrosive environment. A preferred non-frangible thermally responsive fluid control assembly includes a seal assembly, a screw member, and a thermally rated link assembly disposed in a supporting orientation with respect to the sealing assembly to maintain and control transfer of a sealing force of the screw member against the seal assembly in a corrosive environment. The preferred non-frangible thermal actuation fluid control assembly and the preferred relative orientations and geometric relationships between the components minimize and more preferably eliminates any corrosion bridging between the sprinkler components that would otherwise prevent ejection of the seal assembly and/or collapse of the transfer member(s) upon fusion of the thermally rated material. Moreover, preferred embodiments of the fluid control assembly include components having material properties that contribute to the corrosion resistance of the assembly independent of and in conjunction with the relative orientations and geometric relationships. Accordingly, the preferred non-frangible thermally responsive fluid control assembly can maintain seal integrity and proper thermal responsiveness in an extreme salt environment.

A preferred embodiment of a corrosion resistant sprinkler includes a sprinkler frame body having an inlet and an outlet with a passageway disposed therebetween along a sprinkler longitudinal axis. A pair of frame arms preferably extend from the body and converging toward the sprinkler axis to

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form a sprinkler boss with a screw member that is engaged with the sprinkler boss. A fluid deflecting member is coupled to the sprinkler boss. In preferred embodiments of the sprinkler assembly, the frame body and deflecting member are preferably fabricated from a self-passivating material such as, for example, stainless steel. The screw member can also be fabricated from stainless steel, but more preferably is fabricated from a corrosion resistant non-self-passivating alloy, such as for example, a nickel alloy that does not rely on the presence of oxygen to maintain or generate its corrosion resistant surface coating.

A preferred seal assembly is disposed within the outlet that includes an elongated seat member disposed in the outlet. The elongated seat member includes a cylindrical portion that extends into the internal passageway of the sprinkler frame body and a flanged portion is disposed adjacent and outside the outlet of the sprinkler frame body. The flanged portion has a face that preferably includes an elongated slot defined by a slot width and a slot length extending the width of the face. A preferred link assembly is disposed internally with respect to the frame arms to support the seal assembly within the outlet by transferring a compressive sealing force generated preferably by the screw member to the seal assembly.

The preferred link assembly includes a trigger member and a support member joined to one another by thermally rated fusible material. In the preferred embodiment, the trigger member has a solder engagement portion and a canopy or lever portion extending perpendicular to the solder engagement portion. In a preferred embodiment of the trigger member, the lever portion includes a receptacle formed in the first surface for receipt of the leading tip portion of the screw or other load member to define an annular gap therebetween. Moreover, preferred embodiments of the trigger member include a preferred connector formed between the lever and solder engagement portions. The support member is soldered to the solder engagement portion of the trigger member. A preferred embodiment of the support member includes an elongated indentation defining a preferred geometry for housing the solder. The support member has a preferred first edge engaged with the elongated slot of the elongated member of the seal assembly and a second edge located adjacent to and more preferably in contact with the lever portion of the trigger member. The first edge of the support member has a linear length preferably equal to the slot length of the seat member. Moreover, the first edge of the support member preferably defines a thickness that is less than the elongated slot width of the seat member.

The preferred seal, screw member and link assemblies provide exemplary means for non-frangible thermally responsive fluid control that is resistant to corrosive environments. Preferred embodiments of the corrosion resistant sprinkler have been shown to support and maintain the fluid tight seal of the corrosion resistant sprinkler after exposure in an extreme salt spray test for more than ten days and preferably for at least thirty days. Moreover, the preferred corrosion resistant sprinkler has been shown to provide for the desired thermal and actuation response by appropriately actuating in a heated operating test bath. Accordingly, preferred means provide for non-frangible thermal actuation control after exposure to an extreme salt environment. Preferred methods of qualifying a fire protection sprinkler for corrosion resistance and corrosion resistant fire protection are also provided. Additionally, the sprinklers provide a preferred method of corrosion resistant fire protection which

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includes obtaining a sprinkler qualified for corrosion resistance and distributing the sprinkler for installation in a corrosive environment.

BRIEF DESCRIPTION OF DRAWINGS

The accompanying drawings, which are incorporated herein and constitute part of this specification, illustrate exemplary embodiments of the invention, and together, with the general description given above and the detailed description given below, serve to explain the features of the invention. It should be understood that the preferred embodiments are some examples of the invention as provided by the appended claims.

FIGS. 1-2 are various perspective views of a preferred embodiment of a corrosion resistant sprinkler.

FIGS. 3, 3A & 3B are cross-sectional views of the sprinkler of FIG. 1.

FIGS. 4, 4A-4D are various views of a preferred seal assembly for use in the FIG. 1.

FIG. 5 is a perspective view of a preferred non-frangible thermally responsive link assembly for use in the sprinkler of FIG. 1.

FIGS. 5A-5C are various views of a preferred trigger member for use in the link assembly of FIG. 5.

FIGS. 6, 6A & 6B are various views of a support member for use in the link assembly of FIG. 5.

MODE(S) FOR CARRYING OUT THE INVENTION

FIGS. 1, 2 and 3 show various illustrative embodiments of a preferred automatic fire protection sprinkler 10 that includes a sprinkler frame 11 having a body 12 with an inlet 14 and an outlet 16 with an internal passageway 15 extending between the inlet 14 and the outlet 16 along a sprinkler longitudinal axis A--A. Preferably extending from the body 12 are a pair of frame arms 18a, 18b that converge toward the sprinkler axis A--A to form a sprinkler boss 20. A fluid deflecting member 22 spaced from the outlet 16 is preferably coupled to the sprinkler boss 20 for the distribution of firefighting fluid discharged from the outlet 16 upon thermal actuation of the sprinkler 10 in order to address a fire. The sprinkler 10 and its deflecting member 22 are shown in a pendent orientation, but it should be understood that the sprinkler 10 can be configured with an appropriate deflecting member for an upright orientation. The sprinkler 10 is preferably embodied as a corrosion resistant sprinkler in which the sprinkler frame 11 and deflecting member 22 are preferably formed or fabricated from a corrosion-resistant material, such as for example, solid stainless steel, such as for example, UNS-J92800, UNS-S32205, or UNS-S31600. The sprinkler frame 11 and deflecting member 22 are preferably subject to a post-fabrication passivation process per ASTM A 967 to enhance the inherent corrosion resistance of the stainless steel.

The automatic sprinkler 10 includes a preferred means for non-frangible thermally responsive fluid control suitable for use in an environment that can be as corrosive as an extreme salt environment. Generally, the preferred means includes a seal assembly disposed in the outlet of the sprinkler frame and a non-frangible thermally responsive link assembly that controls transfer of a compressive force to the seal assembly to control fluid discharge from the sprinkler outlet. More preferably, the preferred means includes a seal assembly disposed in the outlet of the sprinkler frame, a screw member engaged with the sprinkler frame to generate the

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compressive sealing force and a non-frangible thermally responsive link assembly that controls transfer of the compressive force to the seal assembly to control fluid discharge from the sprinkler outlet. In a preferred embodiment of the means, the link is a multi-member assembly that is preferably held together by a thermally rated fusible material, such as for example a eutectic solder that maintains one or more members of the link assembly in a supporting orientation to directly or indirectly transfer the sealing force to the seal assembly. The supporting orientation minimizes and more preferably eliminates any corrosion bridging that would prevent ejection of the seal assembly and/or collapse of the link member(s) upon melting of the solder. Additionally or alternatively, the screw member is preferably made of a material to minimize and more preferably eliminate corrosive effects on the engagement between the screw member and the link assembly which would otherwise impair proper sealing force generation and transfer. In particular embodiments, the screw member is fabricated from a corrosion resistant material that is non-self-passivating. As used herein, a "non-self-passivating" material is a material that does not require reaction between the base material and the surrounding atmosphere to create its corrosion resistant surface coating. An exemplary non-self-passivating material is nickel alloy.

The preferred means can maintain seal integrity in an extreme salt environment and thus can provide the desired non-frangible thermally responsive fluid control in corrosive conditions. As described herein, the preferred means has been shown to support and maintain the fluid tight seal of the corrosion resistant sprinkler after exposure to an extreme salt environment over a significant period of time, e.g. more than ten days and more preferably at least 30 days. In addition, after the extreme salt exposure, the preferred means has been shown to appropriately thermally respond to a heated ambient temperature by collapsing and separating from the seal assembly. Thus, the preferred means can maintain a fluid tight seal and provide for the desired thermal and actuation response despite exposure to such corrosive conditions. Although, the preferred means described herein provide for a corrosive resistant non-frangible means for use with the corrosion resistant sprinkler frame and deflector, the preferred means can alternatively be used in a corrosion proof sprinkler assembly in which the sprinkler frame and/or deflector is coated with special coatings, plating, or finish. Further in the alternative, preferred embodiments of the non-frangible means can be incorporated into a standard brass sprinkler frame.

A preferred embodiment of the non-frangible means includes a seal assembly 50 disposed within the outlet 16, a fusible thermally responsive link assembly 100 in contact with the seal assembly 50 to define the preferred supporting orientation with respect to one another and a preferred screw member 80 engaged with the link assembly 100 to transfer a compressive sealing force against the seal assembly 50. In response to a sufficiently high ambient temperature, the link assembly 100 thermally fuses, collapses, separates from the seal assembly 50 and the seal assembly 50 ejects from the outlet 16 under internal fluid pressure. Fluid discharged from the outlet 16 impacts the deflector member 22 to address a fire.

The inlet 14, internal passageway 15 and outlet 16 define the sprinkler orifice and its discharge characteristics of the sprinkler. As is known in the art, the discharge characteristics of a sprinkler is determined by its K-factor, which is defined by $K=Q/P^{1/2}$, where Q represents the flow rate (in gallons/min (GPM)) of water from the outlet of the internal

passage through the sprinkler body and P represents the pressure (in pounds per square inch (psi)) of water or firefighting fluid fed into the inlet end of the internal passageway through the sprinkler body. The discharge characteristics of the preferred sprinkler body 12 define a nominal K-factor in a range of 5.6 to 28.0 GPM/(psi)^{1/2} [80 to 400 L/min/(bar)^{1/2}]. More particularly, preferred embodiments of the sprinkler and sprinkler body for use in the sprinkler 10 defines a nominal K-factor which range from 5.6 to 14.0 GPM/(psi)^{1/2} [80 to 200 L/min/(bar)^{1/2}] and are yet even more preferably any one of 5.6, 8.0, or 11.2 GPM/(psi)^{1/2} [80, 115, or 160 L/min/(bar)^{1/2}] and are even more preferably 8.0 GPM/(psi) 12 [115 L/min/(bar)^{1/2}].

As seen in FIGS. 3A and 3B, the preferred link assembly 100 is held together by a thermally rated solder to support a first end 100a of the link assembly 100 in engagement within a recess of the seal assembly 50 disposed in the outlet 16 and to support a second end 100b of the link assembly having a receptacle 110 formed therein for engagement with the compression or load screw 80 to transfer the compressive force from the screw 80 to the seal assembly 50. The engagement between the first end 100a of the link assembly and the seal assembly 50 defines a preferred geometry in which the recess surface area has a ratio of exposed area-to-area in contact with the first end 100a of the link assembly which preferably ranges from 2:1 to 2.5:1. In one preferred embodiment, a minimum surface area of contact between the first end 100a of the link and the seal assembly is preferably 0.007 square inches to form a fluid tight seal. The preferred ratio minimizes and more preferably eliminates corrosion build-up that may bridge between the link assembly 100 and the seal assembly 50 that would otherwise interfere with sprinkler actuation and operation upon melting of the solder in the appropriate ambient temperature and the subsequent ejection of the seat assembly 50 under the internal fluid pressure delivered to the sprinkler.

Additionally or alternatively, the engagement between the second end 100b of the link and the load screw 80 defines another preferred geometry to define a preferred annular crevasse or gap about the load screw 80. In a preferred aspect, a minimum surface area of contact between the second end 100b of the link assembly and the load screw 80 is sufficient to form the engagement between the components while minimizing and more preferably eliminating corrosion build-up that may bridge between the link assembly 100 and the load screw 80 that might interfere with proper sealing function in the unactuated state of the sprinkler assembly or otherwise interfere with the proper sprinkler actuation and operation upon melting or fusion of the solder. In preferred embodiments described herein, the receptacle 110 defines a maximum depth and the engagement defines a maximum width of the annular gap between the screw 80 and the link 80, which together define a preferred ratio of maximum gap width-to-maximum receptacle depth that ranges from 2:1 to 3:1. The screw member 80 is preferably fabricated from a corrosion resistant non-self-passivating material. Thus, to the extent a corrosion build-up forms at the engagement site between the link assembly 100 and the screw member 80 thereby depriving the site of oxygen for self-passivation, the non-self-passivating property of the preferred screw member 80 can provide corrosion resistance to the assembly. Similarly, for embodiments of the sprinkler assembly 10 that do not include an annular gap between the link 100 and the screw member 80 to expose the screw to atmosphere, a preferred

non-self-passivating screw member 80 can provide corrosion resistance to the assembly at the second end 110b of the link assembly.

Preferred embodiments of the sprinkler assembly 10 were subjected to a series of corrosion resistance tests, as described herein, including a corrosion salt spray test and an operating temperature liquid bath test performed preferably in accordance with one or more of industry accepted standards, such as for example in accordance with any one of FM Approval Standards FM 2000, FM 2008, FM 2030 or UL 199. The sprinkler assemblies passed the corrosive resistance tests thereby demonstrating their suitability for installation and use in a corrosive environment. Accordingly, by testing the preferred corrosion resistant sprinkler assembly and testing the sprinkler in an appropriate corrosion test, preferred methods for qualifying an automatic sprinkler for corrosion resistance are provided herein.

In one preferred corrosion test, preferred embodiments of a sprinkler were subjected to a corrosion salt spray test performed in accordance with FM 2000. A test group of as many as eight sprinklers are hydrostatically tested to confirm that there is no weep or leak points at or below 500 psi. Each sprinkler inlet was filled with deionized water and sealed with a plastic cap and supported within a test chamber in its intended installation orientation, e.g., pendent. The sprinklers were then exposed to a salt spray fog as specified by ASTM B117, Standard for Salt (Fog) Testing. Chamber conditions monitored daily include: chamber temperature, volume of condensation collected, and specific gravity of solution. The salt solution consisted of twenty percent (20%) by weight of common salt (sodium chloride) dissolved in deionized water. The samples were exposed for a period of over ten days and more preferably exposed for at least thirty days (30 days). After the exposure period, the test sprinklers are dried for a two-to-four day period and visually inspected for any signs of severe deterioration or impending failure. Without any such visual evidence, the test sprinklers were then subjected to a hydrostatic test pressure of 175 psi for one minute without leakage. The test sprinklers successfully passed by the visual and hydrostatic testing thereby demonstrating that the preferred sprinkler assemblies 10 could withstand an extreme salt environment and maintain seal integrity. Post-salt operational testing was conducted on the test sprinklers to show that the sprinkler assemblies still appropriately thermally respond despite the extreme salt exposure.

Following the salt fog exposure test, the test sprinklers were tested for conformance for sensitivity in accordance with FM 2000. The thermal sensitivity of the trigger assembly and sprinkler is characterized by Response Time Index ("RTI"), measured in units of (ft.-s)^{1/2} [(m-s)^{1/2}]. For one or more tested sprinklers, the RTI was calculated based upon the time to sprinkler operation in a plunge test performed in accordance with FM 2000. The calculated RTI was then compared for compliance as falling within an appropriate Standard Response or Quick Response limits as specified by FM 2000.

According to the plunge test procedure, the test sprinklers are mounted within a test tunnel and placed under pneumatic pressure. The sprinkler and its thermally responsive trigger are exposed to a stream of gas at a given velocity and temperature specified under FM 2000 for the nominal temperature rating of the test sprinkler. Once exposed to the test gas stream, the time to sprinkler operation is monitored and determined. Generally, the RTI calculation is a function of: (i) the determined sprinkler operation time, (ii) the test gas temperature, and (iii) the nominal operating temperature and

(iv) the thermal conductivity (C-factor) of the test sprinkler, which is defined as the measure of conductance between the sprinkler's heat responsive element and the other components of the sprinkler as measured in $(\text{m/s})^{1/2}$ [$(\text{ft/s})^{1/2}$]. Under FM 2000, the RTI of a "Standard Response Sprinkler" is between $80 (\text{m-s})^{1/2}$ and $350 (\text{m-s})^{1/2}$ [between $145 (\text{ft-s})^{1/2}$ and $635 (\text{ft-s})^{1/2}$] with a C-factor equal to or less than $2.0 (\text{m/s})^{1/2}$ [$3.62 [(\text{ft/s})^{1/2}]$]. For a "Quick Response Sprinkler," the calculated RTI is equal to or less than $50 (\text{m-s})^{1/2}$ [$90 (\text{ft-s})^{1/2}$] with a C-factor equal to or less than $1.0 (\text{m/s})^{1/2}$ [$1.81 [(\text{ft/s})^{1/2}]$].

Preferred embodiments of the corrosion resistant sprinkler with their non-frangible link assemblies **100** are configured as Standard Response Sprinklers and complied accordingly in the thermal sensitivity plunge test after satisfactorily passing the 30-day salt fog exposure test. One preferred embodiment of sprinkler has a nominal temperature rating of 161°F . with a C-factor of $1.57 (\text{ft/s})^{1/2}$ to provide an RTI of about $185 (\text{ft-s})^{1/2}$ and more preferably $185.86 (\text{ft-s})^{1/2}$. Another preferred embodiment of sprinkler has a nominal temperature rating of 205°F . with a C-factor of $2.39 (\text{ft/s})^{1/2}$ to provide an RTI of about $215 (\text{ft-s})^{1/2}$ and more preferably $214.8 (\text{ft-s})^{1/2}$. By satisfying the RTI testing requirements, the preferred corrosion resistant sprinklers demonstrated that the preferred sprinkler assemblies **10** could maintain conformance with the appropriate thermal response despite the long-term exposure to the extreme salt conditions.

In another post-salt fog exposure test, the test sprinklers were subject to an operating temperature (liquid bath) test performed in accordance with FM 2000, which illustrates a typical test set-up. The operating test verify that the sprinkler actually operates within a preferred percentage of its marked nominal temperature rating when immersed in constant rate-of-temperature-rise liquid bath. More specifically under the standardized tests, for test sprinklers having a nominal temperature rating of less than 400°F ., the sprinkler is to demonstrate an actual operating temperature within 3.5% of its nominal temperature rating. For test sprinklers having a nominal temperature rating of over 400°F ., the sprinkler is to demonstrate an actual operating temperature within 7% of its nominal temperature rating. In accordance with the standardized test procedures, the sprinklers are placed in a test vessel on grate or rack above the bottom of the vessel. The vessel is filled with a liquid bath corresponding to the nominal rating of the sprinkler: (i) water for a nominal sprinkler temperature of $0-175^\circ \text{F}$.; (ii) glycerin for a nominal sprinkler temperature of $176-360^\circ \text{F}$.; and (iii) vegetable oil for a nominal sprinkler temperature over 361°F . The test sprinklers were immersed in the liquid.

In accordance with the test, the bath is raised to an initial test temperature within twenty degrees Fahrenheit ($20^\circ \text{F}/11^\circ \text{C}$.) of the temperature rating of the sprinkler. The test bath is then raised at a constant rate, preferably ranging from $0.5-0.8^\circ \text{F}$. per minute until the sprinkler is operated or until the bath temperature is twenty degrees Fahrenheit ($20^\circ \text{F}/11^\circ \text{C}$.) above the rated temperature of the test sprinklers. The test sprinklers satisfactorily passed the operating temperature test by operating as intended with the requisite percentage of their rated temperature. Accordingly, the preferred sprinkler assemblies **10** and their preferred seal and link assemblies resisted the corrosive effects of long-term exposure to the extreme salt conditions and properly actuated and separated within the rated temperature of the link assembly.

The seal assembly **50** and link assembly **100** define a preferred geometric engagement. With reference to FIG. 3,

the preferred seal assembly **50** includes an elongated seat member **52** with a resilient seal **54**, such as for example a Belleville spring, disposed about the seat member **52** to form a fluid tight seal with an internally formed sealing surface of the sprinkler body **12**. The resilient seal **54** is preferably nickel alloy, encapsulated in PTFE and coated on both sides with PTFE tape. The preferred elongated seat member **52** includes a cylindrical portion **56** extending into the internal passageway **18** of the sprinkler frame body and a flanged portion **58** disposed adjacent and outside the outlet **16** of the sprinkler frame body **12**. The cylindrical portion **56** preferably includes an asymmetrical chamfer **57** to facilitate ejection of the seal assembly upon actuation of the preferred non-frangible fusible link **100**.

Referring to FIG. 3A, the flanged portion **58** has a face post **60** opposed to the sprinkler boss **20**. The face post **60** preferably includes an elongated slot **62** defining the preferred recess of the seal assembly **50** and extending perpendicular to the sprinkler axis A--A in the direction or parallel to the plane P1 defined by the frame arms **18a**, **18b** and perpendicular to a bisecting plane P2 that intersects the plane P1 of the frame arms along the sprinkler axis A--A. The preferred link assembly **100** engages the elongated slot **62** to define the preferred orientation between the components to support or maintain the seal assembly **50** within the outlet **16** in the unactuated state of the sprinkler **10** for proper sprinkler operation in a corrosive atmosphere and more particularly extreme salt environment. Shown in FIGS. 4, 4A-4D are various views of the preferred seal assembly **50**. The face post **60** is preferably a circular cylindrical formation with the elongated slot **62** preferably defined by a slot width SW and a slot length SL in which the slot length SL preferably extends the diameter of the circular face post **60**. Alternatively, the face post **60** can be non-circular, where instead the elongated slot extends the entire width of the face **60**. Moreover, where the elongated member **52** defines a total axial seat height STH, the slot length SL defines a preferred seat height-to slot length ratio STH:SL that ranges from 1:1 to 2:1. As seen in FIGS. 4B and 4D, the elongated slot **62** is defined by a slot depth SD which is preferably constant over its entire slot length SL. The slot depth SD can also be substantially constant over the slot width SW as seen in FIG. 4D or alternatively increase toward the slot center. The elongated slot **62** of the seat member **52** is configured to engage the link assembly **100** in the preferred manner as described herein. More particularly, the elongated slot **62** defines a sufficient area for contact with the link assembly **100** with sufficient space therebetween to minimize or eliminate corrosion bridging which may otherwise interfere with the proper separation of the components upon fusion of the link assembly. In preferred embodiments of the elongated member **52**, the slot depth SD preferably ranges from about 0.01 in. to 0.025 in. and more preferably ranges from 0.012 to about 0.022 inch.

Referring again to FIGS. 1, 2 and 3 the preferred fusible link assembly **100** is disposed interiorly between the frame arms **18a**, **18b** and loaded against the seal assembly **50** by the screw member **80** that is threadedly engaged in the sprinkler boss **20** to generate a compressive force against the link assembly **100**. The loading screw **80** includes an engagement tip **82** for engaging the link assembly **100** and a tool engagement end **84** to thread the loading screw **80** into the sprinkler boss **20**. In preferred embodiments of the sprinkler **10** described herein, the load screw **80** and link assembly **100** also define a preferred geometric interaction that maintains the seal assembly **50** in a fluid tight sealed engagement with the internal sealing surface of the sprinkler

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body **12** against the force of the resilient seal **54** and any fluid pressure acting against the seal assembly **50**. For the preferably corrosion resistant sprinkler, the screw member **80** is preferably constructed from 18-8 stainless steel.

Referring again to FIG. 3A, the preferred link assembly **100** is preferably a two-component assembly that is fused together by a thermally rated amount of solder. The preferred link assembly **100** includes a first trigger member **102** and a second support member **150** soldered to one another by a preferred volume of solder located or housed within and surrounding an indentation **152** formed in the support member **150**. A preferred solder is a eutectic solder for 205° F. (96° C.) or alternatively a non-eutectic solder for 161° F. (72° C.). Each trigger member **102** and support member **150** are preferably constructed from beryllium nickel, such as for example, UNS-N03360 beryllium nickel. The components are preferably subject to one or more finishing coatings and more preferably a single finishing coating. In one illustrative two-coat finish, the components are finished with one coat of a primer to a thickness ranging from 0.7-1.2 MIL and another coat of a lacquer to a thickness ranging from 0.3-0.8 MIL.

The trigger member **102** includes a solder engagement portion **104** and a canopy or lever portion **106** that is coupled to and substantially skewed with respect to the solder engagement portion **104**. More preferably, the canopy portion **106** extends perpendicularly with respect to the solder engagement portion **104**. The canopy portion **106** also preferably extends so as to intersect the sprinkler longitudinal axis A--A. The canopy portion **106** has a first surface **108a** opposed to the sprinkler boss **20** for engagement with the load screw member **80**. The canopy portion **106** also has a second surface **108b** opposite the first surface **108a** and opposed to the flanged portion **58** of the elongated member **52** of the seal assembly **50** for a preferred contact with the support member **150**.

The support member **150** has a first surface **154a** in which the indentation **152** is formed for soldered engagement with the solder engagement portion **104** of the trigger member **102**. An exposed second surface **154b** of the support member shows the indentation **152** as a preferred linearly formed relief or projection **152** on the exposed surface **154b**. As seen in the sprinkler assembly in FIG. 3A, the support member **150** has a first edge **156** for engagement with the elongated slot **62** of the seal assembly **50**. A second edge **158** of the support member **150** is located adjacent the second surface **108b** of the lever portion **106** of the trigger member **102**. More preferably, the second edge **158** is in supporting contact with the lever portion **106** of the trigger **102** to define a pivot P about which the trigger member **102** rotates upon thermal actuation of the trigger.

Schematically shown are various forces acting on the trigger member **102** in the unactuated state. The first surface **108a** includes the receptacle **110** formed in the first surface **108a** for receipt of the leading tip portion **82** of the load screw member **80**. The receptacle **110** locates application of the compression force F-screw, or a component thereof, from the screw member **80** at a distance X1 from the pivot P in a direction perpendicular the direction of force F-screw. Accordingly, the force applied by the screw member **80** applies a moment about the pivot P that acts to separate the link components **102**, **150**. Resisting the separating force is the reactive or resulting force F-Tens of the solder which acts at a distance Y1 from the pivot P in a direction perpendicular to the force F-Tens. Accordingly, the force F-Tens applies a counter-moment to that generated by the screw member **80** to maintain the link assembly in the

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unactuated state of the trigger. Upon fusion of the solder, the force F-Tens is reduced and the moment applied by the screw member **80** separates the components **102**, **150** resulting in the collapse of the link assembly **100**.

In preferred embodiments of the sprinkler **10** and its link assembly, the preferred distance X1 between the pivot and application of the load force F-screw can enhance the corrosion resistance performance of the sprinkler. In particular, the preferred arrangements can facilitate satisfaction of the 30-day salt spray fog and thermal sensitivity tests previously described. Accordingly, the moment generated by load F-Screw at the preferred distance X1 can overcome extreme salt corrosive effects that may otherwise interfere with the proper operation of the link assembly upon fusion of the solder. Moreover, the moment generated by load F-screw and distance X1 is of a sufficient size that the solder can maintain the link assembly **100** together with a force F-Tens in the unactuated state of the sprinkler for proper thermal response and in particular for proper standard response compliance in the thermal sensitivity test.

In one preferred embodiment of the link assembly **100**, the receptacle **110** is configured to apply the load F-Screw at distance X1 ranging from 0.015-0.020 inch and preferably 0.018 inch from the pivot P. A preferred load F-Screw to form a fluid tight seal with the seal assembly **50** ranges from about 150-175 lbs-force. Resolving for the static force F-Tens required by the solder to maintain the assembly **100**, the force F-Tens is applied at a preferred distance of about 0.4 inch and more preferably 0.38 inch from the pivot P and is preferably no less than 4 lbs-force and is more preferably about 8 lbs.-force. Accordingly, the link and the solder in an unactuated state preferably can withstand at least 8-lbs in tension without failing while providing the preferred thermal sensitivity for proper operation. Preferred embodiments of the non-frangible soldered assembly are configured for a preferred maximum load in tension of about 25 lbs.-force. Because the preferred 8-lbs. force at the solder from the applied moment of the load force is less than the maximum design load of the link assembly, the preferred link can maintain the preferred geometry to seal the sprinkler in an unactuated state and separate upon fusion of the solder material.

Shown in FIG. 3B is a detailed cross-sectional view of a preferred engagement between the receptacle **110** of the link assembly **100** and the screw tip **82** of load screw **80**. As shown, the preferred receptacle **110** is a concave formation in the first surface **108a** of the lever portion **106** that defines a maximum depth of the receptacle **110**. In the engagement of the screw tip **82** within the receptacle **110**, the screw tip **82** preferably contacts the bottom of the concave receptacle **110** and forms an annular gap **113** about the screw tip **82**. In the embodiment shown, the annular gap **113** is at its maximum width at the surface **108a** of the lever portion **106**. Formation of the annular gap **113** between the load screw **80** and the receptacle is preferably defined by the difference in geometry between the receptacle **110** and the screw tip **82**. In a preferred embodiment of the assembly forming a gap **113**, the receptacle **110** defines a first radius of curvature R1 that is greater than a second radius of curvature R2 defined by the load screw **80**. In alternate embodiments of the sprinkler assembly, the receptacle **110** can define a first radius of curvature equal to the second radius of curvature R2 of the load screw **80** such that there is no gap formation between the surface of the receptacle **110** and the screw tip **82**. Accordingly, depending upon the receptacle **110** con-

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figuration, the first radius of curvature R1 and second radius of curvature R2 define a ratio R1:R2 that ranges from 1:1 to 1.25:1.

In an alternate embodiment (not shown), the receptacle **110** can be formed as a through hole in the lever portion **106** that extends from the first surface **108a** through to the second surface **108b**. Upon load screw **82** engagement, the screw tip **82** preferably extends through the lever portion **106** and is exposed to the surrounding atmosphere. The exposure can facilitate corrosion resistance where the load screw **82** is made of a self-passivating material. Depending upon the configuration of the through hole receptacle, the receptacle can be of a constant width, taper narrowly in the direction of increasing depth, or broaden in the direction of increasing receptacle depth. Accordingly, depending upon the configuration of the through hole receptacle, there can be a gap between the load screw and the through hole surface or none at all.

Alternatively or additionally, the receptacle **110** is defined by a perimeter geometry that circumscribes the engaged screw tip **82** at the first surface **108a** of the lever portion **106**. In another preferred aspect, the perimeter circumference can define a preferred ratio with the circumference of the screw tip **82** at the first surface **106a** that ranges from 1:1 to 1.2:1. The perimeter of the receptacle **110** at the first surface **108a** can be circular or non-circular, such as for example rectangular, oval, triangular or otherwise a non-symmetrical closed form. In an embodiment in which the perimeter is circular, the diameter of the receptacle perimeter defines a preferred ratio with the diameter of the engaged screw at the first surface **108a** that ranges from 1:1 to 1.2:1.

The gap **113** is sufficiently large to minimize and more preferably eliminate any corrosion build-up between the screw tip **82** and the lever portion **106** of the link assembly **100** and thereby contribute to the corrosion resistance of the sprinkler assembly. Moreover, by minimizing the corrosion build-up at the site of the screw and link engagement, the surface of the load screw **82** can remain exposed to the surrounding atmosphere. Thus, in an embodiment in which the load screw **82** is fabricated from a self-passivating material such as, for example, a stainless steel alloy, the corrosion resistant coating of the load screw can be generated and/or maintained to contribute to the corrosion resistance of the sprinkler assembly.

Notwithstanding the ability of the preferred geometric relationship between the screw **80** and the link **100** to resist corrosion, the load screw **80** is preferably alternatively fabricated from a non-self-passivating corrosion resistant material. A preferred non-self-passivating material for fabrication of the load screw **80** is nickel alloy. Thus, to the extent any corrosion build-up forms between the screw tip **82** and the lever portion **106** of the link assembly **100**, the non-self-passivating corrosion resistant coating of the screw member **80** is not negatively impacted by the deprivation of oxygen.

Shown in FIGS. 6, 6A and 6B is the support member **150**. Each of the first and second edges **156**, **158** has a linear portion **156a**, **158a** that extends perpendicular to a bisecting axis B--B. The linear portions **156a**, **156** extends perpendicular to the sprinkler longitudinal axis A--A in the sprinkler assembly **10**. The linear portion **156a** of the first edge **156** is preferably engaged with the elongated slot **62** and has an edge length EL1 equal to the slot length SL of the elongated slot **60** of the elongated member **52**. Moreover, the thickness TH of the support member **150** at the linear portion **156a** of the first edge **156** is preferably smaller than the slot width SW of the elongated slot **62** of the seal assembly **50**

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and more preferably defines a slot width-to-support edge thickness ratio SW:TH that ranges from 2:1 to 2.5:1. Accordingly, for the preferred support and seat seal assembly engagement, the support edge **156** has sufficient surface contact to transfer the sealing force from the screw **80**. Additionally, the wider slot width provides a gap about the support edge thickness so as to minimize and more preferably eliminate corrosion bridging between the support and the seat member, which may otherwise interfere with their proper separation. In preferred embodiments of the link assembly, each of the trigger and support members **102**, **150** are of a material thickness ranging from 0.025-0.030 inch and more preferably 0.027 inch thick. Individually and in their assembly, the components provide sufficient strength to the link assembly **100** under the various loads to maintain the fluid-tight seal in the unactuated state of the sprinkler.

The linear portions **156a**, **158a** of the first and second edge **156**, **158** are spaced apart along the bisecting axis B--B to define the maximum length L1 of the support member. The elongated indentation **152** is formed between the first and second edge **156**, **158** and preferably extends parallel to and centered about the bisecting axis B--B. The preferably linear indentation **152** has a first end portion **153a** and a second end portion **153b** with the indentation **152** having a preferred constant width W from the first end portion **153a** to the second end portion **152b** such the indentation **152** is preferably symmetrical about a second axis C--C that is perpendicular to the first axis B--B. Accordingly, the preferred indentation **152** is formed to include a pair of spaced apart lateral linear edges **155a**, **155b** extending parallel to the bisecting axis B--B. In a preferred embodiment, the indentation **152** is vertically off center such that the second end portion **153b** of the indentation **152** defines a first distance to the first edge **156** of the support member **150** and the first end portion **153a** of the indentation **152** defines a second distance to the second edge **158** of the support member that is different and preferably larger than the first distance. The indentation **152** defines an indentation length L2 and more preferably defines a support member-to-indentation length ratio L1:L2 that ranges from 1.1:1 to 1.4:1. In alternate embodiments, the indentation **152** can define alternative configurations which is asymmetrical about the second axis C--C. The indentation **152** can, for example, have variable widths along its length to define alternate geometries, such as for example, T-shaped or cross-shaped.

The support member **150** has a first lateral edge **160** and a second lateral edge **162** extending parallel to one another and the bisecting axis B--B to define the maximum width W1 of the support member **150**. The first and second edges **156**, **158** of the support member **150** extend between the first and second lateral edges **160**, **162**. The length EL1 of the linear portion **156a** of the first edge **156** and the maximum width W1 of the support member **150** define a preferred width-to-length ratio W1:EL1 ranging from 1.5:1 to 2:1. In a preferred embodiment, the linear portion **156a** of the first edge **156** is centered between the first and second lateral edges **160**, **162**. The first edge **156** includes a pair of transition portions **156b**, **156c** disposed about the central linear portion **156a**. The transition portions **156b**, **156c** extending from the central linear portion **156a** to one of the first and second lateral edges **160**, **162**. Each transition portion **156b**, **156c** is preferably linear and skewed with respect to the central linear portion **156a** to define an acute transition angle α , which is preferably about twenty degrees (20°).

In another preferred aspect, the linear portion **158a** of the second edge **158** includes two linear portions **158a'**, **158a''**

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with an arcuate portion **158b** centered between the two linear portions. The second edge **158** also preferably includes a pair of transition portions **158c**, **158d** disposed about the two linear portions **158a**. Each transition portion **158c**, **158d** extend from the two linear portions to one of the first and second lateral edges **156**, **158**. Each transition portion **158c**, **158d** is preferably linear and skewed with respect to the two linear portions to define an acute transition angle β , which preferably ranges from 65-70 degrees and is more preferably sixty-eight degrees (68°). Given, the transition portions at each of the first and second edges **156**, **158** of the support member **150**, the first and second lateral edges **160**, **162** of the support member **150** are preferably smaller than the pair of lateral edges **153a**, **153b** of the indentation **152**.

As seen in FIGS. **5A**, **5B**, the first and second surfaces **108a**, **108b** of the canopy portion **106** are preferably parallel to one another to define the canopy portion **106** as a substantially planar member and more preferably a polygonal planar member. The canopy portion **106** includes a free leading end **112** having a first width WW1 and an opposite trailing end **114** contiguous with the solder engagement portion having a second width WW2. The first width WW1 is preferably different than the second width WW2 and more preferably is less than the second width WW2.

The trigger member **102** preferably includes a connector portion **116** formed between the solder engagement portion **104** and the opposite trailing end **114** of the canopy portion **106**. Laterally, the connector portion **116** preferably locates the solder engagement portion **104** closer to the opposite trailing end **114** of the canopy portion **106** than the free leading end **112**. With reference to FIG. **5C**, the connector portion **116** includes a first planar portion **118a** and a second planar portion **118b**.

Shown in FIG. **3A** is a detailed view of the link assembly **100** within the sprinkler frame. The cross-section of the interior channel **120** preferably defines a centroid C disposed preferably to a side of the solder engagement portion **104** opposite the line of action of the load screw, which is substantially aligned with the sprinkler axis A--A. Accordingly in the preferred sprinkler assembly **10**, the solder engagement portion **104** of the trigger **102** is centered between the line of action of the load screw **80** and the centroid C. More preferably, the centroid C and the line of action are spaced apart by a distance of about 0.1 inch. Alternatively, or additionally, the interior channel is configured so as to approximately locate the centroid C closer to the lever or canopy portion **106** than the support member **150**. In a preferred embodiment the centroid C is approximated as being at the intersection of the mid-line of the channel height CH and a line between and parallel to the support member **150** and the first planar portion **118a**. More preferably, the approximate centroid C defines a preferred ratio of distances to support member (C-150)-to-distance to canopy portion (C-106) that ranges from 1:1 to 2.5:1 and is more preferably about 1.2:1.

Referring again to FIG. **5C**, the solder engagement portion **104** includes a first end **105** contiguous with the connector **116** and an opposite terminal end **107**. The free leading canopy end **112** of the canopy portion **106** is axially spaced from the terminal end **107** of the solder engagement portion **104** to define a height HH of the trigger **102** to define a trigger height-to-channel height ratio (HH:CH) that ranges from 2:1 to 3:1. The height HH of the trigger **102** and its soldered engagement with the support member **150** keeps the terminal end **107** of the trigger **102** out of the elongated slot **62** of the seal assembly **50**. To facilitate appropriate

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engagement with and separation from the support member **150**, the interior surface of the solder engagement portion **104** preferably includes one or more dimples or projections for alignment with the preferred indentation **152** of the support member and one or more receptacle disposed about the dimples.

The preferred fusible link assembly **100** provides for a preferred two-piece non-frangible thermally responsive assembly. Other two-piece fusible link assemblies can be used with the preferred seal assembly provided the link assembly defines the preferred engagement geometry with the seal assembly to transfer the sealing force from the compressive screw member while minimizing or eliminating corrosion build-up therebetween to provide a corrosion resistant non-frangible thermally responsive fluid control.

In other alternate embodiments, the non-frangible thermally responsive link assembly can be alternatively embodied in an assembly having more than two components. For example, instead of the support and trigger members being directly held together by the solder material, the assembly can incorporate a fastener (not shown) which holds the support and trigger member in an assembled relationship to transfer the sealing force to the seal assembly. The fastener can be held in place by the solder material such that upon fusion of the solder material, the fastener separates from the support and trigger members thereby permitting the link assembly to collapse and the sprinkler is actuated. In another alternative embodiment, the support and trigger members can be configured and positioned for transfer of the sealing force indirectly to the seal assembly. For example, the support and trigger members can be joined by the solder material to instead hold a strut and lever in an assembled orientation which transfers the compressive force of the screw member to the seal assembly. In such an arrangement, upon fusion of the solder material and separation of the support and trigger members, the lever member and strut members collapse and the sprinkler is actuated.

The preferred embodiments of sprinkler assembly and extreme salt corrosion testing provide preferred methods of qualifying a sprinkler for corrosion resistance and providing corrosion resistant fire protection in which the sprinkler has a non-frangible thermally responsive fluid control assembly. A preferred method of corrosion resistant fire protection includes obtaining a fire protection sprinkler having a non-frangible thermally responsive fluid control assembly that is qualified for resistance to an extreme salt environment by corrosion testing the sprinkler in a manner as described herein. Obtaining a fire protection sprinkler can include any one of manufacturing, acquiring and/or purchasing the sprinkler. The method further preferably includes distributing the sprinkler for installation in a corrosive environment, which preferably includes giving, supplying, selling or otherwise providing the sprinkler for installation and use in a corrosive environment application.

While the present invention has been disclosed with reference to certain embodiments, numerous modifications, alterations, and changes to the described embodiments are possible without departing from the sphere and scope of the present invention, as defined in the appended claims. Accordingly, it is intended that the present invention not be limited to the described embodiments, but that it has the full scope defined by the language of the following claims, and equivalents thereof.

What is claimed is:

1. A corrosion resistant fire protection sprinkler comprising:

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a sprinkler frame body of corrosion resistant material having an inlet and an outlet with a passageway disposed therebetween along a sprinkler longitudinal axis; a fluid deflecting member spaced from the outlet and coupled to a sprinkler boss;

a seal assembly disposed within the outlet; and

a link assembly means for non-frangible thermal actuation control after exposure to an extreme salt environment, wherein the link assembly means include a trigger member and a support member contacting the trigger member,

wherein the trigger member comprises a connector portion between a first portion and a second portion, the first portion contacting the support member to define a pivot point, the second portion having a terminal end disposed between the seal assembly and the sprinkler boss and confronting the seal assembly, and the connector portion defining a centroid to a side of the first portion, and

wherein (i) the pivot point is offset from the sprinkler longitudinal axis, and/or (ii) the trigger member and the support member are compressed between the sprinkler boss and the seal assembly, and/or (iii) the trigger member is positioned to, upon melting of a fusible, temperature responsive solder, rotate about the pivot point away from the support member due to a moment generated by a compressive force applied to the seal assembly, and/or (iv) the support member and the trigger member extend along a link axis non-parallel to the sprinkler longitudinal axis,

wherein the means, after thirty days of exposure to a salt spray fog, maintains a fluid tight seal at the outlet, and

wherein the means has a thermal sensitivity and after the thirty days of exposure to the salt spray fog, thermally actuates in accordance with its thermal sensitivity, and/or

wherein the means has a normal operating temperature and after the thirty days of exposure to the salt spray fog, thermally actuates in a heated bath test raised to the normal operating temperature.

2. The sprinkler of claim 1, wherein the seal assembly includes a recess positioned to receive and contact the support member, and wherein a ratio of an area of the recess exposed to an area of the recess in contact with the support member is between 2:1 and 2.5:1.

3. The sprinkler of claim 1, further comprising:

a load screw in threaded engagement with the sprinkler boss, wherein a lever portion of the trigger member includes a receptacle to receive and contact the load screw, and

wherein a ratio of a width of a gap between the load screw and the receptacle to a depth of the receptacle is between 2:1 and 3:1.

4. The sprinkler of claim 1, wherein the link axis intersects the sprinkler longitudinal axis at a point at which the support member contacts the seal assembly.

5. The sprinkler of claim 1, further comprising a load screw positioned to apply a compressive force to the seal assembly through a lever portion of the trigger member and the support member.

6. The sprinkler of claim 1, further comprising:

a load screw comprising a non-self-passivating screw in threaded engagement with the sprinkler boss to generate a compressive force against the trigger member and the support member.

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7. The sprinkler of claim 1, further comprising a load screw, wherein a lever portion of the trigger member includes a receptacle or a through hole positioned to receive the load screw, and wherein the load screw comprises a self-passivating screw.

8. The sprinkler of claim 1, further comprising a load screw, wherein a lever portion of the trigger member includes a receptacle to receive and contact the load screw, wherein the receptacle has a first curvature defined by a first radius, wherein a portion of the load screw contacting the receptacle has a second curvature defined by a second radius, and wherein a ratio of the first radius to the second radius is between 1:1 and 1.25:1.

9. The sprinkler of claim 1, wherein the link assembly means further include a solder material that directly holds the trigger member to the support member.

10. The sprinkler of claim 1, wherein the link assembly means further include a fastener coupling the trigger member to the support member.

11. The sprinkler of claim 1, wherein the link assembly means further includes a solder material that directly holds or a fastener that holds the trigger member to the support member.

12. The sprinkler of claim 1, wherein (i) the pivot point is offset from the sprinkler longitudinal axis, and (ii) the trigger member and the support member are compressed between the sprinkler boss and the seal assembly.

13. The sprinkler of claim 1, wherein (i) the trigger member is positioned to, upon melting of the fusible, temperature responsive solder, rotate about the pivot point away from the support member due to a moment generated by a compressive force applied to the seal assembly, and (ii) the support member and the trigger member extend along the link axis non-parallel to the sprinkler longitudinal axis.

14. The sprinkler of claim 1, wherein each of the sprinkler frame body and the fluid deflecting member consists of stainless steel.

15. The sprinkler of claim 1, wherein the sprinkler frame body includes a wrench boss configuration.

16. The sprinkler of claim 1, further comprising:

a load screw in threaded engagement with the sprinkler boss,

wherein the trigger member includes a solder engagement portion and a lever portion extending from the solder engagement portion, and

wherein the link assembly means define a centroid disposed to a side of the solder engagement portion opposite to a line of action of the load screw.

17. The sprinkler of claim 1, wherein the trigger member includes a solder engagement portion and a lever portion extending from the solder engagement portion, and wherein the lever portion intersects the sprinkler longitudinal axis.

18. The sprinkler of claim 1, wherein the trigger member includes a lever portion, and wherein the support member includes a first edge configured to engage with the seal assembly and a second edge in supporting contact with the lever portion to define the pivot point.

19. The sprinkler of claim 1, wherein the sprinkler defines a nominal K-factor ranging from 5.6 to 14.0 GPM/(psi)^{1/2}.

20. The sprinkler of claim 1, wherein the trigger member includes a solder engagement portion and a lever portion extending from the solder engagement portion, wherein the lever portion is positioned to press toward the sprinkler boss.

21. The sprinkler of claim 20, wherein the support member is positioned to press toward the seal assembly, and wherein the support member remains in contact with the

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trigger member by a fusible, temperature responsive solder at the solder engagement portion.

22. The sprinkler of claim 1, wherein the support member contacts the seal assembly.

23. The sprinkler of claim 22, wherein the seal assembly includes an elongated slot, and wherein the support member includes a first edge configured to engage the elongated slot of the seal assembly.

24. The sprinkler of claim 1, wherein each of the trigger member and the support member consists of a nickel alloy.

25. The sprinkler of claim 24, wherein the nickel alloy consists of beryllium nickel.

26. The sprinkler of claim 1, further comprising a pair of frame arms extending from the sprinkler frame body and converging toward the sprinkler longitudinal axis.

27. The sprinkler of claim 26, wherein the pair of frame arms converge toward the sprinkler longitudinal axis to form a sprinkler boss, and wherein the fluid deflecting member is coupled to the sprinkler boss.

28. A fire protection sprinkler comprising:

a stainless steel sprinkler frame body having an inlet and an outlet with a passageway disposed therebetween along a sprinkler longitudinal axis;

a stainless steel fluid deflecting member spaced from the outlet and coupled to a sprinkler boss;

a seal assembly disposed within the outlet; and

a link assembly means for non-frangible thermal actuation control after exposure to an extreme salt environment, wherein the link assembly means include a nickel alloy trigger member and a nickel alloy support member contacting the trigger member, wherein:

the trigger member comprises a connector portion between a first portion and a second portion, the first portion contacting the support member to define a pivot point, the second portion having a terminal end between the seal assembly and the sprinkler boss and confronting the seal assembly, and the connector portion defining a centroid to a side of the first portion, and the pivot point is offset from the sprinkler longitudinal axis, and/or the trigger member and the support member are compressed between the sprinkler boss and the seal assembly, and/or

the trigger member is positioned to, upon melting of a fusible temperature responsive solder, rotate about the pivot point away from the support member due to a moment generated by a compressive force applied to the seal assembly, and/or

the support member and the trigger member extend along a link axis non-parallel to the sprinkler longitudinal axis,

wherein:

the means, after thirty days of exposure to a salt spray fog, maintains a fluid tight seal at the outlet, and

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the means has a thermal sensitivity and, after the thirty days of exposure to the salt spray fog, thermally actuates in accordance with its thermal sensitivity, and/or

the means has a normal operating temperature and, after the thirty days of exposure to the salt spray fog, thermally actuates in a heated bath test raised to the normal operating temperature.

29. The sprinkler of claim 28, wherein:

the pivot point is offset from the sprinkler longitudinal axis, and the trigger member and the support member are compressed between the sprinkler boss and the seal assembly, and

the trigger member is positioned to, upon melting of the fusible, temperature responsive solder, rotate about the pivot point away from the support member due to a moment generated by a compressive force applied to the seal assembly, and

the support member and the trigger member extend along the link axis non-parallel to the sprinkler longitudinal axis.

30. The sprinkler of claim 28, wherein each of the trigger member and the support member consists of a nickel alloy.

31. The sprinkler of claim 28, wherein each of the sprinkler frame body and the fluid deflecting member consists of stainless steel.

32. The sprinkler of claim 28, wherein the sprinkler frame body includes a wrench boss configuration.

33. The sprinkler of claim 28, further comprising:

a load screw in threaded engagement with the sprinkler boss,

wherein the trigger member includes a solder engagement portion and a lever portion extending from the solder engagement portion, and

wherein the link assembly means define a centroid disposed to a side of the solder engagement portion opposite to a line of action of the load screw.

34. The sprinkler of claim 28, wherein the trigger member includes a solder engagement portion and a lever portion extending from the solder engagement portion, and wherein the lever portion intersects the sprinkler longitudinal axis.

35. The sprinkler of claim 28, wherein the trigger member includes a lever portion, and wherein the support member includes a first edge configured to engage with the seal assembly and a second edge in supporting contact with the lever portion to define the pivot point.

36. The sprinkler of claim 28, wherein the sprinkler defines a nominal K-factor of 5.6 or 8.0 GPM/(psi)^{1/2}.

37. The sprinkler of claim 28, further comprising a pair of frame arms extending from the sprinkler frame body and converging toward the sprinkler longitudinal axis.

38. The sprinkler of claim 37, wherein the pair of frame arms converge toward the sprinkler longitudinal axis to form a sprinkler boss, and wherein the fluid deflecting member is coupled to the sprinkler boss.

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